NITIN HARISHCHANDRA SHUKLA

Cybersecurity Researcher · Network Security & Tooling · SOC / Penetration Testing

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PROFESSIONAL SUMMARY

Cybersecurity professional pursuing an M.Sc. in Cyber Security at the University of Mumbai, building on a B.Sc. in Computer Science. Published researcher in network-protocol security diagnostics and the author of a substantial portfolio of security tooling in Rust, Kotlin and Python — spanning packet analysis, QUIC/RFC 9000 auditing, non-root traffic capture, OSINT reconnaissance and web-security assessment. Hands-on with ethical hacking, vulnerability assessment and threat analysis, seeking a role as a SOC Analyst, Penetration Tester or Cybersecurity Analyst.

EDUCATION

M.Sc. in Cyber Security 2025 – 2027 (Expected)

University of Mumbai, Mumbai · SGPA 8.0 / 10.0 (First Year)

B.Sc. in Computer Science Jun 2022 – May 2025

Thakur Ramnarayan College of Arts & Commerce, University of Mumbai · CGPA 8.93 / 10.0

RESEARCH & PUBLICATION

NetworkSentinel — Cross-Platform Network Reconnaissance & QUIC Diagnostics **PEER-REVIEWED** Published 2026

Rust · Nmap/Wireshark-style scanning & packet capture · Npcap / libpcap · QUIC (RFC 9000) · Windows & Android

- Designed a memory-safe, cross-platform Rust framework unifying active host/service scanning, live packet capture, protocol decoding, bidirectional flow tracking and PCAP post-mortem analysis in a single CLI toolkit.
- Engineered a dedicated QUIC diagnostic subsystem implementing all five RFC 9000 security-relevant compliance tests (anti-amplification limit, Initial-packet padding, version negotiation, 0-RTT anti-replay, path validation) — a capability absent from Nmap and Wireshark.
- Compiled one shared Rust core for Windows (Npcap) and Android (Termux), producing identical diagnostic output, including an A+ (100/100) QUIC audit grade, on both platforms.

Publication: *N. H. Shukla, "NetworkSentinel: An Integrated Cross-Platform Framework for Network Reconnaissance, Packet Analysis, and QUIC Diagnostics," International Journal of Emerging Technologies and Innovative Research (JETIR), ISSN 2349-5162, Vol. 13, Issue 9, pp. c179–c183, Sept 2026.* View Paper

SECURITY PROJECTS

Packetra — Android Network-Visibility & Security-Assessment Toolkit

Kotlin · Jetpack Compose · Hilt · Room · VpnService · OkHttp

- Built a defensive-first, non-root packet analyzer that captures the device's own traffic via VpnService, with per-app attribution, offline ASN/GeoIP enrichment, DPI protocol classification and TLS-SNI / HTTP-host extraction, plus an on-device firewall and PCAPng / PCAP-over-IP / HAR export.
- Added non-destructive web-security assessment (security headers, CSP, cookie flags, CORS, TLS telemetry, tech fingerprinting, secret scanning) with CVSS 3.1 triage and JSON / SARIF 2.1.0 / Markdown reporting, and passive OSINT reconnaissance (Certificate Transparency, DoH, RDAP/WHOIS, SPF/DMARC/DKIM, subdomain-takeover checks).
- Enforced a deny-by-default engagement-scope validator (GET-only, SSRF-range blocking, bounded concurrency) so every active check stays authorized and non-exploitative.

N3-Core — Offline Notebook, Code Editor & PCAP Reader

Kotlin · Jetpack Compose · Room · pure-Kotlin PCAP parser

- Developed a fully on-device Android app pairing a VS Code-style multi-language code editor and syntax checker with an advanced PCAP / PCAPNG reader that decodes Ethernet, IPv4/IPv6, TCP/UDP, DNS, HTTP, TLS (SNI) and QUIC, with follow-stream, a combined IP/port/transport filter and text export.
- Engineered a virtualized packet list and large-file editor that stay responsive across hundreds of thousands of packets and 32 MB files; secured notes with per-item AES-256-GCM encryption keyed in the Android Keystore.

Injectra — Privacy-Focused Android Browser & Web-Testing Workbench

Kotlin · Android WebView (Chromium) · OkHttp

- Built a hardened WebView browser with process-isolated private windows, a ~200-network ad/tracker blocklist, HTTPS upgrading and a locked-down WebView attack surface (no JS bridges, blocked file/content/javascript navigations, sanitized intent handling).
- Implemented a developer-tools and Burp-style testing suite — request/response header and rule engine (block, redirect, response rewrite, JS/CSS injection), network inspector, request repeater, encoder/decoder (JWT, hashing), passive scanner and scope control — explicitly scoped to the browser's own traffic with a published capability matrix of what is and isn't in scope.

SessionGram — Multi-Account Telegram Client

Python · Telethon · aiohttp · Kotlin / WebView (Android) · Chaquopy

- Created a multi-account client keeping 150+ Telegram accounts connected simultaneously behind one interface, with phone+OTP and 2FA login, bulk session import, a unified inbox and a SHA-256 passcode lock.
- Optimised for scale with bounded connection pools and virtualized rendering, cutting worst-case UI frame time from ~230 ms to ~60 ms under a 600-message burst; keeps all session secrets on-device with no server, analytics or tracking.

Phishing Detection & Automated Takedown System

Python · URL/domain analysis · WHOIS · DNS reconnaissance · SMTP automation

- Built a phishing-detection system analysing URL structure, IP-based domains and WHOIS registration metadata to flag malicious sites, feeding an automated, evidence-backed abuse-reporting pipeline routed to hosting-provider and registrar abuse contacts.

Additional Tools

- **DorkMaster (Android, Kotlin)** — a Google-dork search-link generator across 20 categories for manual OSINT review (link generation only, no automated crawling). **Internship Finder (Python, Streamlit)** — a personalised bulk-outreach web app deployed live on Streamlit Cloud.

CERTIFICATIONS & VIRTUAL INTERNSHIPS

Cybersecurity Virtual Internship — AICTE EduSkills (Palo Alto Networks) Jul – Sep 2025

National Internship Portal, Ministry of Education · 10 weeks · Grade B

Cyber Security Internship (AICTE) — AISECT Learn 2025

4-credit AICTE internship

Job Simulations (Forage): AIG Cybersecurity Analyst · Telstra Cybersecurity · Mastercard Cybersecurity · Deloitte Australia Cyber

Certifications: SQL Injection Attacks — EC-Council · Cyber Security & Ethical Hacking — Quantum Learnings · CCNA Networking — self-trained (Udemy)

TECHNICAL SKILLS

Security Domains: Ethical Hacking, Penetration Testing, Vulnerability Assessment, OSINT, Malware Analysis, Network Security, Packet Analysis, Phishing Detection & Threat Analysis, QUIC / Transport-Protocol Security

Tools: Nmap, Wireshark, Burp Suite, Metasploit, Npcap / libpcap, Shodan, VirusTotal, crt.sh, DNSDumpster, WHOIS

Programming: Rust (systems / CLI tooling), Kotlin (Android / Jetpack Compose), Python, Scripting & automation

Platforms & Frameworks: Android (VpnService, WebView, Room, Hilt), Telethon, Streamlit, Cargo workspaces, GitHub, CI/CD workflows

LANGUAGES

English (Proficient) · Hindi (Native)